

Household Donut Chart -- Implementation Plan

What's being replaced

The current Members card (flat list with a per-member progress bar) plus the MemberSheet portal (spending breakdown, role editor) are being replaced by a two-level animated donut that lives directly in the Household tab. A new HouseholdDonutChart.tsx component is the main deliverable. Household.tsx is the only other file that changes structurally.

Phase 1 -- Static Household Donut

New file: artifacts/finance-app/src/components/HouseholdDonutChart.tsx

Modified: artifacts/finance-app/src/pages/Household.tsx

Chart-building logic (buildHouseholdChart)

The SVG coordinate system is cloned verbatim from DonutBudgetChart (CX=160, CY=160, RI=75, RO=128, EXPAND=14, CAT_GAP=2.5). Instead of categories, the input is the members array plus the household budget.

Each member becomes a group:

Spent arc -- memberColor, fraction = spentInViewerCurrency / effectiveBudget

Remaining arc -- memberColor darkened 52% (same hexDarken helper), fraction = remainingBudget / effectiveBudget

Over-budget -- single arc at memberColor with red glow filter (same isOverBudget path as DD)

effectiveBudget is householdBudget when set, otherwise sum(memberBudgets) (proportional fallback). If householdBudget > sum(memberBudgets), an Additional Funds group is appended as a neutral gray arc sized to the difference. Members whose totalBudget is null contribute only a spent arc (no remain), treated as budget-0 for group sizing.

All member amounts are converted to the viewer's currency using the rates prop already available in Household.tsx.

Interactions (Phase 1 only -- no drill-down yet)

Single tap on a segment -> invoke that member: segment expands outward by EXPAND units, member name + spent + % shown in SVG center text (same mechanism as DD's selectedCat). Tap again -> deselect.

Double-tap center -> compact ? expanded mode toggle, same as DD (legend slides in/out via max-width / margin-left CSS transition).

Legend on the right: member avatar initial (or ? for virtual), name, spent amount. Dimming of non-selected items when one is invoked.

Animations inherited from DonutBudgetChart

Pulse hint (3 firings: 3 s, 8 s, 13 s) -- same donutBlink037/045/053/061 keyframes, already injected globally. Fires in compact mode only.

Wiggle hint (4 s after mount) -- two-beat "detach" on groups 1 and 4, same Web Animations API pattern. Guarded by disableAnimations pref and checkDonutWiggleDue().

Red glow filter (feGaussianBlur + feColorMatrix) -- injected per-instance, applied when spent > budget && budget > 0.

Household.tsx wiring

The `{/* ?? Members ?? */}` card (currently lines 1437-1511) is replaced by `<HouseholdDonutChart members={members} householdBudget={budgetInViewerCurrency} currency={prefs.currency} rates={splitRates} onMemberInvoked={setInvokedMember} />`.

The member count badge, invite button, and the head-admin functions currently inside the flat list are moved to a small "Members" header row immediately above the donut (count + invite button for head), keeping visual parity.

Phase 2 -- Personal Donut Drill-Down & Lock Animation

Modified: HouseholdDonutChart.tsx, Household.tsx

State machine inside HouseholdDonutChart

drillPhase: "idle" | "fade-others" | "expand-arc" | "personal" | "collapse-arc" | "restore-others"

drilledMemberId: number | null

personalData: SpendingItem[] | null

isPrivate: boolean

Trigger

Double-tapping the already-invoked member's segment (i.e., selectedMemberId !== null && doubletap on that segment OR the SVG center while a member is selected) triggers drill-down. The lastSegTapRef per-member pattern (same 350 ms window as DD's lastCenterTapRef) tracks this.

Animation sequence -- Household -> Personal (?1.1 s total)

Step	Duration	What happens
A -- fade-others	350 ms	All groups except the invoked member animate opacity -> 0 via CSS t
B -- expand-arc	500 ms	The invoked arc's angular span interpolates from its current [startDeg,
C -- cross-dissolve	300 ms	The household <svg> fades to opacity 0; the <DonutBudgetChart> wr

Implementation note: Step B is achieved by keeping arcProgress: 0->1 in a ref updated by rAF. On each frame, the arc path is recomputed.

Personal view

Once drillPhase === "personal":

Render a standard <DonutBudgetChart spending={personalData} totalBudget={member.totalBudget} currency={viewerCurrency} hasData={true} /> in place of the household SVG.

A back-hint appears in the expanded center text: "double tap to exit" (reuses existing t("donut.xx_to_exit") key).

Double-tap the personal donut's center -> reverse sequence: dissolve back, contract arc, restore others. drillPhase returns to "idle", selectedMemberId stays set so the member remains visually invoked.

Data fetching

useGetMemberSpending(drilledMemberId) is enabled only when drillPhase !== "idle". The data arrives during the animation's Steps A+B (350+500 ms = 850 ms window), which is plenty of time for the API call. A spinner appears in the dissolving household SVG center if data hasn't arrived by Step C.

Private dashboard -- lock animation

If the fetch returns a 403 / error, isPrivate = true. Instead of showing a personal donut:

Step C resolves but the <DonutBudgetChart> is replaced by a plain SVG circle (same RI/RO dimensions, all segments gray).

In the SVG center, a <g> renders a padlock icon path (inline SVG, no external font). It plays a CSS keyframe:

0 ms: scale(0), opacity 0

150 ms: scale(1.25), opacity 1

250 ms: scale(1), opacity 1 -- hold

800 ms & 1200 ms: two donutBlink045 / donutBlink053 circle pulses (amber/orange tint, not gray) fire around the padlock using the

existing hint circle mechanism

2000 ms: padlock opacity -> 0 (fade out 400 ms)

After 2500 ms: static text renders in the center: member name + t("hh.private_lock_msg") ("Private dashboard")

Double-tap anywhere in the gray donut -> collapse back to household view (same reverse sequence).

Head exception: head users never get a 403, so the lock animation never fires for them.

Phase 3 -- Household Spendings Virtual Member

Modified: HouseholdDonutChart.tsx

Drill-down for userId === ?1

The virtual "Household Spendings" member (amber, ? icon) triggers the same animation sequence as Phase 2. No lock -- always accessible.

Data source

```
GET /households/members/household-spendings/spending (already built). Each item has isRecurringPayment: true, total (0 if unpaid, amount if paid), budget (the RP amount), categoryColor (RP color).
```

SpendingItem mapping

```
applied RP -> { total: rp.amount, budget: rp.amount, isRecurringApplied: true, isLarderDesignated: false }
pending RP -> { total: 0, budget: rp.amount, isRecurringApplied: false, isLarderDesignated: false }
```

These map cleanly into DonutBudgetChart's existing logic -- applied ones render a full colored arc, pending ones render only the dimmed remain arc. The existing ? badge in the legend (isRecurringApplied && ?) already handles the tick display.

totalBudget passed to DonutBudgetChart = sum of all household RP amounts.

Phase 4 -- Admin Preservation, Cleanup, Edge Cases

Modified: Household.tsx, HouseholdDonutChart.tsx

Head admin functions

MemberSheet currently handles role editing and member removal. These are preserved via a "Manage Members" collapsible section placed below the donut, visible only to heads. It renders the same flat list (smaller, denser) with a ?? tap per member that opens the existing MemberSheet -- but with its spending-breakdown section hidden (replaced by the donut drill-down). The MemberSheet receives a new hideSpending: boolean prop to suppress that section.

Alternatively (simpler): a ... icon overlaid on the member's legend row in expanded mode, head-only, opens MemberSheet with hideSpending. This avoids any extra card.

18n keys to add

Key	English
hh.additional_funds	Additional Funds
hh.household_donut_label	Household Spending
hh.private_lock_msg	Private dashboard
hh.drill_back_hint	double-tap to return

Edge cases

Scenario	Handling
Member has totalBudget = null	Spent arc only; excluded from "additional funds" calculation; budget = 0 in legend
All members have 0 budget AND householdBudget not set	Show equal-sized segments per member (effectiveBudget = totalSpent ? 1)
householdBudget = null, sum of member budgets = 0	Empty state: no donut, "No budgets set" message
Member has spent = 0, budget = null	Render as a 1/N minimum-width placeholder segment so they appear in the chart
disableAnimations = true	All requestAnimationFrame loops, hint timers, wiggle timers skip; mode transitions are instant
Personal data loads after animation completes	Spinner stays until data arrives, then DonutBudgetChart replaces it
Rates not yet loaded	Convert 1:1 (same as current MemberSheet behaviour), re-render once rates arrive

Cleanup

Remove old member list card from Household.tsx (lines 1437-1511)

Remove memberAnchorY state, setMemberAnchorY, and the MemberSheet invocation at line 1868 if the management-only version is wired through the new manage-members collapsible instead

HouseholdDonutChart owns its own data-fetching for drill-down (via enabled queries), so no new props are needed for spending data in Household.tsx

File change summary

File	Change
src/components/HouseholdDonutChart.tsx	New -- all four phases built here
src/pages/Household.tsx	Remove member list card, add <HouseholdDonutChart>, preserve admin
src/pages/Dashboard.tsx	No change
src/components/DonutBudgetChart.tsx	No change (used as-is for personal drill-down)
i18n JSON files	Add 4 new keys

Ready to build whenever you say go.